

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **Single Line Diagrams (SLDs)** and detailed **wiring diagrams** for LV electrical switchgear panels, ensuring adherence to **IEC 61439** standards for over 15 distinct panel configurations.
- Performed critical electrical calculations including **load calculations**, **short-circuit analysis**, and **busbar sizing**, resulting in optimized component selection and enhanced system safety for custom switchgear designs.
- Selected and specified a range of electrical components such as **circuit breakers**, **busbars**, **protection relays**, and **contactors**, balancing performance requirements with cost-effectiveness for industrial applications.
- Coordinated closely with **mechanical design teams** and **manufacturing teams** to integrate electrical schematics into panel layouts, reducing design iteration cycles by approximately 20%.
- Prepared comprehensive **technical documentation**, including detailed **Bill of Materials (BOMs)** and **test procedures**, supporting the smooth transition from design to production for enterprise switchgear projects.
- Contributed to **factory acceptance tests (FAT)** and site commissioning phases, troubleshooting design-related issues and ensuring deployed systems met performance specifications and client expectations.

SKILLS

Electrical Design – Switchgear design, LV systems, Power distribution systems, Schematics, Wiring diagrams, Layout drawings, Panel arrangements, Single Line Diagrams (SLDs)

Component Selection – Circuit breakers, Busbars, Protection relays, Contactors, Control devices

Standards & Compliance – IEC 61439, IEC 60439, NSAI IS10101:2025

Electrical Calculations – Load calculations, Short-circuit analysis, Thermal analysis, Busbar sizing

Design Tools – AutoCAD Electrical, EPLAN

Documentation – Bill of Materials (BOM), Test procedures, Operation & maintenance manuals

Project Support – Manufacturing support, Factory Acceptance Tests (FAT), Site commissioning, Design troubleshooting

PROJECTS

LV Switchgear Panel Design for Industrial Facility (HCL Technologies)

- Developed detailed **Single Line Diagrams (SLDs)** and **schematics** for a large-scale LV switchgear panel, intended for a major industrial manufacturing plant.
- Conducted thorough **load calculations** and **short-circuit analysis** to ensure component compatibility and system integrity under peak operational demands.
- Generated a complete **Bill of Materials (BOM)** and draft **test procedures**, coordinating with the production team to validate the design against manufacturing capabilities.

Custom Switchgear Component Selection & Integration (HCL Technologies)

- Evaluated and selected specialized **protection relays** and **control devices** from multiple vendors (e.g., Siemens, ABB, Schneider) for a custom switchgear solution.
- Created detailed **wiring diagrams** that integrated selected components with existing power distribution infrastructure, ensuring seamless system connectivity.
- Collaborated with operations teams to document the integration process, reducing post-installation troubleshooting time for similar custom panels by an estimated 15%.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)